

DIGITAL PRODUCT DESIGN

Lecture 8: A/B Testing and Experiments

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Outline

- A/B Testing
- Potential Pitfalls
- Examples

A/B TESTING OVERVIEW

Correlations and Causation

- Office 365 users that see more error messages are less likely to churn (i.e. retention is higher).
 - Should Microsoft degrade their code to reduce churn?
- Experiments are the gold standard for establishing causal relationships
 - An experiment that reduces errors would show a decrease in churn

Motivation for Experiments

- Hard to assess the value of an idea ex ante
 - Do customers spend more if the website is blue or black?
- Hard to quantify causal impact of changes / ideas on key metrics
 - How does a new feature impact engagement, sales, OEC
 - How does advertising impact bottom line?
- Experiments allow us to test many ideas and quantify causal impact

Motivation for Experiments, Continued

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Source: Kohavi, Tang, and Xu

Hard to Assess New Ideas

- Google tested 41 shades of blue
- Head of design thought it was crazy
- Turned out to matter!
 - Engagement increased substantially



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About

For over 150 years, generations of Boilermakers have left their ...

Quantify Impact

- Website latency impacts user experience
 - Users like fast websites
- Engineers are expensive!
- How to determine if it is worthwhile to hire an engineer to reduce latency?
- Microsoft has run many such experiments
 - Found that 10 millisecond decrease in latency pays for an engineer
 - Takes ~200 milliseconds to blink an eye

RUNNING A/B TESTS

Choosing the Unit of Randomization

- Determine what you will randomize into the different treatment arms
 - User, user-session, page, day, store, ...
- Decision balances:
 - Feasibility: Can we randomize at this level?
 - E.g. hard for Walmart to randomize at customer level in its stores
 - Power
 - Randomizing at most granular level usually good for statistical power
 - Interference concerns
 - Do units interact with each other?

Randomize then Analyze

- Randomly split traffic between two (or more) versions
 - Version A: Control, often the status quo
 - Version B: Treatment, includes some changes
- Collect metrics of interest
- Test for differences between the groups
 - We call the difference the **estimated treatment effect**
- Question: Is assigning the first 500 customers to treatment and the next 500 to control a good way to randomize? Why not?
 - Randomization does not mean arbitrary assignment

Analyze Differences

- Estimate the difference in average outcomes among groups
 - Estimate the difference $\tau = E[Y|D = 1] - E[Y|D = 0]$
 - D = Treatment dummy variable
 - τ is called the treatment effect
- To test if τ is significantly different from 0, can estimate this using a regression
 - A/B testing platforms will handle this for you)

Statistical Hypothesis Testing

- We want to know how likely the difference between treated and control would occur by chance
 - Typically involves calculating p-values and confidence intervals
- Hypothesis testing gives us a way to do this
 - Null hypothesis: Treatment effect = 0
 - Alternative: Treatment effect is not zero
- p-value: Probability under the null hypothesis of observing an absolute treatment effect at least as large as what we observed

Analyze Differences - Heterogeneous Treatment Effects

- Heterogeneous treatment effect: How the treatment impacts groups differently
- Estimate the treatment effect among sub-groups and compare
 - You can also nest this in a single regression
- Example: Estimate treatment effect among mobile users and desktop users

POTENTIAL PITFALLS

When Experiments Fail

- Randomization fails
- Interference
- Statistical problems
- Ethics and experimentation

Randomization Fails

- Actually randomizing users can be hard
 - A/A Tests are useful to test the experiment pipeline
- Easy to test for mistakes implementing randomization
- Two useful tests to run
 - Sample ratio mismatch: Test that the share of users receiving each treatment equal the expected shares
 - Balance tests: Test that the treated and control groups have similar pre-treatment characteristics

Sample Ratio Mismatch

- Sample ratio mismatch: Test that the share of users receiving each treatment equal the expected shares

- Estimate a regression

$$D_i = \beta_0 + \varepsilon_i$$

- D_i : A dummy equal to 1 if user i is treated
- Test that β_0 equals expected share of treated users
 - Default is not correct p-value, it tests the null that $\beta_0 = 0$!

Balance Tests

- Balance tests: Test that the treated and control groups have similar pre-treatment characteristics
- Estimate the treatment effect of treatment on pre-treatment characteristics
 - If randomization worked, these should be equal

$$Y_i = \beta_0 + \beta_1 D_i + \varepsilon_i$$

- What is Y_i ? What test would we run?

Interference

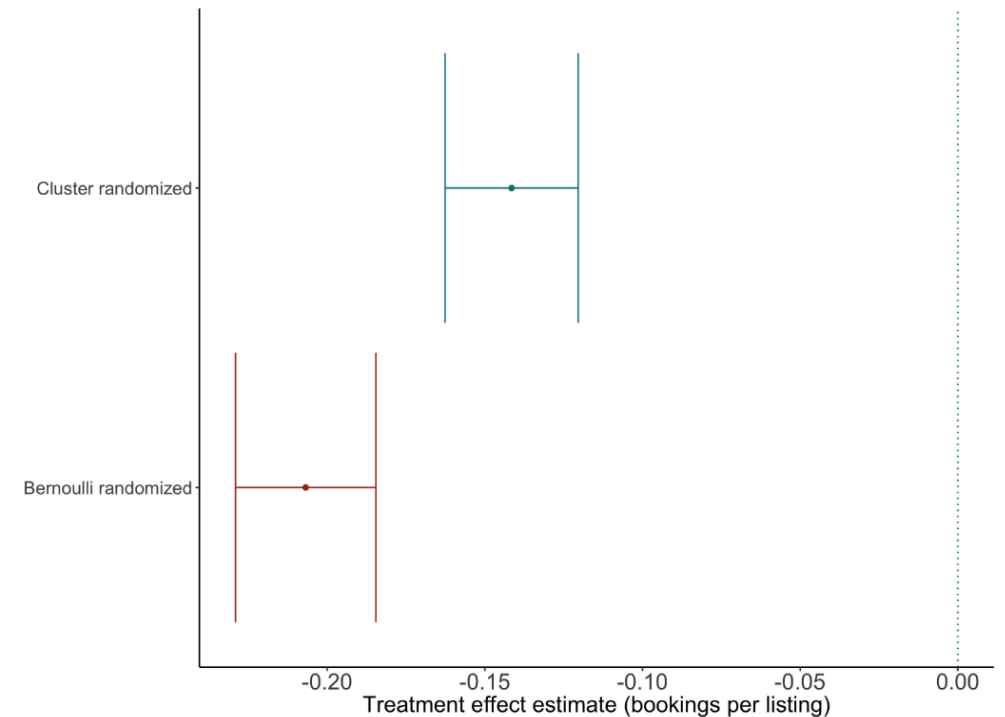
- Interference occurs when units in the experiment interact with one another
- Examples:
 - Marketplaces: If we treat some sellers but not other, do they have a competitive advantage?
 - Networks: If treatment impacts my posting behavior, this impacts control users who interact with me
- Traditional statistical analysis is incorrect if interference is present

Interference - Solutions

- Randomize at a higher level where interference concerns are less severe
- Examples:
 - Marketplaces: Randomize at the market level (i.e. Etsy product category)
 - Networks: Cluster randomization (i.e. randomize friend groups who frequently interact)

Interference – Holtz et al (2020)

- Airbnb determining how its fee impacts demand
- There is interference as higher fee impacts prices and competition
- Naïve randomization overestimates the elasticity of demand
- Clustering similar units leads to better estimates



Statistical Problems – Multiple Hypothesis Testing

- Under the null hypothesis, 5% of experiments will reject the null at the 5% significance level
- As we run many tests, we will have some false positives
 - Reject a true null hypothesis
- Run an experiment 20 times, we expect at least 1 to show a significant difference
- Particularly an issue with heterogeneous effects.
 - Many subgroups we can look at so likely some will show differences by chance
- Also impacts interpretation of A/B tests
 - Often true effect sizes are smaller than experimental estimates suggest

Statistical Problems – Peeking

- If I look at the data every day, this is a version of the multiple testing problem
- Leads to an elevated false positive rate

Statistical Problems – Solutions

- Solution is to use statistical inference that corrects for multiple comparisons or anytime valid inferences
- Beyond the scope of this course
 - Many testing platforms have this built in
 - If you encounter, know this is an adjustment for these issues

Ethical Experimentation

- Many experiments businesses run involve minimal risk to participants
- There is often limited oversight of experiments in industry
- Establishing a culture of ethical experimentation is important
 - An internal IRB can be helpful in reviewing tests
- Academic experiments regulated by the Belmont report
 - Focuses respect for persons, minimizing risks, and justice (fair distribution of risks / benefits)

Examples of Problematic Experiments

- Facebook promoted negative content to some people to test emotional contagion
- Found that negative content induced people to post more negative content
- Public backlash ensued
 - Lack of consent was important

Facebook Doesn't Understand The Fuss About Its Emotion Manipulation Study

Kashmir Hill Former Staff

Welcome to The Not-So Private Parts where technology & privacy collide

Examples of Problematic Experiments, Continued

- OkCupid randomly altered its match scores it showed users
- Goal was to see if they are predicting or causing relationships
- Public backlash ensued
 - Deception was key concern

OkCupid Lied To Users About Their Compatibility As An Experiment

Kashmir Hill Former Staff

Welcome to The Not-So Private Parts where technology & privacy collide

Ethics in Experiments

- There is active debate on the ethics of such experiments
- Key concern is:
 - Trading off benefits and risks
 - Ensuring participants are treated as autonomous users
- A useful question:
 - Would you deploy this change to 100% of users without testing?
- “Wall street journal” test is also a useful thought experiment

A/B TESTING STRATEGY

A/B Tests Aren't Run in a Vacuum

- If we allocate sample to test one idea, it often takes sample away from testing another idea
- There is a tradeoff between testing many ideas with small sample-sizes vs few ideas with big sample sizes
- How do we decide which is better?
 - “A/B Testing with Fat Tails” by Azevedo et al (2020)
 - The paper is dense

When Few Big Experiments are Optimal

- If ideas are predictable and tend to have marginal improvements best to run fewer larger experiments
- Carefully triage ideas
 - Identify the most promising
- Because effects tend to be smaller, large experiments let us identify the positive-payoff

When Many Small Experiments are Optimal

- When ideas are plentiful and there are “heavy tails” best to run many small experiments
 - This means there are many ideas with outlier payoffs
- The idea is that it is important to find these outlier ideas
 - Hard to identify ex ante
- Because effects are big, they can be found in smaller experiments
- This method will not reliably identify the ideas that have small positive impacts
 - Worth the tradeoff to find the home-run ideas

What is Best for Our Company?

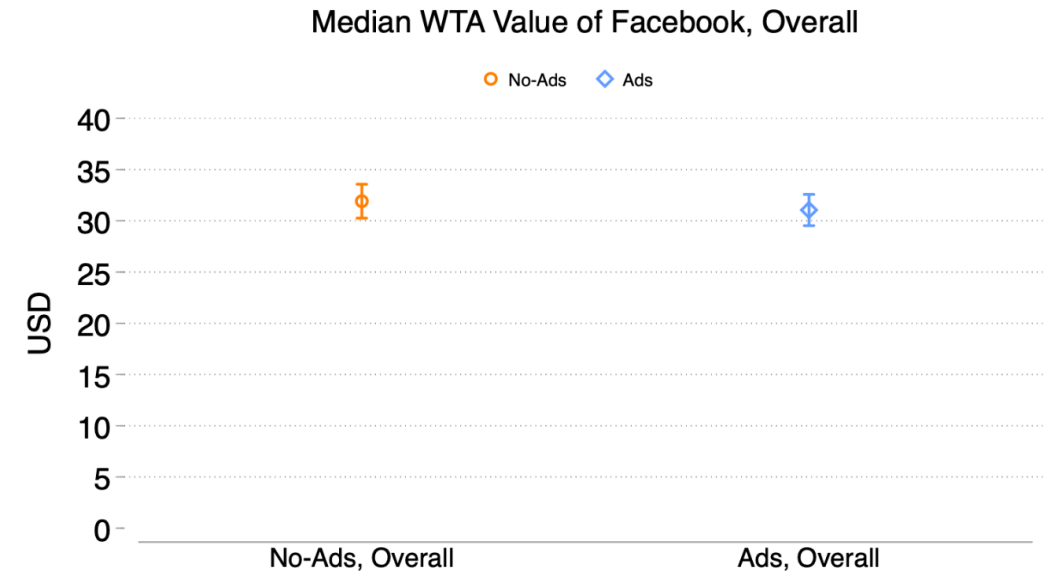
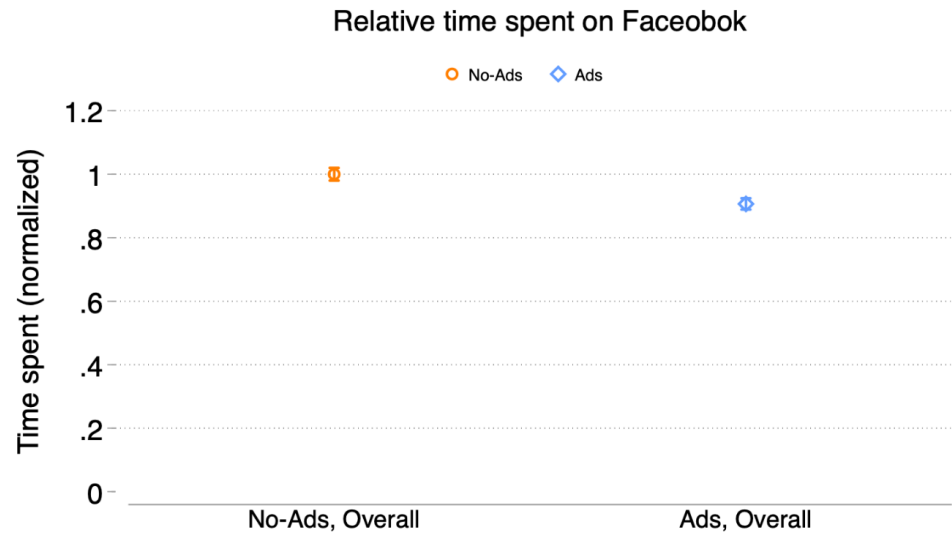
- How easy is it to run experiments?
 - In online world, often relatively easy
- Do ideas have “fat-tailed” payoffs?
 - In new ventures, often yes as there are many low-hanging fruit
 - Surprisingly, often true in big organizations as well
 - Azevedo et al (2020) found Bing had very fat tails

MINI CASE STUDY

What is Best for Our Company? Continued

- Facebook's revenue has been relying heavily on advertising. In early this year, Facebook reported advertising revenue at \$16.6 billion for the final quarter of 2018, up 30 percent year-over-year. Facebook news feed ads allow businesses to run targeted campaigns that appear as posts in Facebook news feeds.
- Imagine yourself as a PM for FB news feed ads team. You are interested in increasing ad-revenue and want to measure user sensitivity to ads.
- Describe an experiment that would allow you to estimate this quantity
 - What challenges do you have to overcome here?

Facebook Ran a Version of Experiment



Source: Brynjolfsson et al (2024)

QUESTIONS?